001: over_simple

$$\frac{1}{2}$$

002: over_grouped

$$\frac{a+b}{c-d}$$

003: over_nested

$$\frac{\frac{1}{2}}{3}$$

004: over_glued_denom

$$\frac{7}{10}$$

005: over_in_paren

$$\left(\frac{x}{y}\right)$$

006: sqrt_ident

$$\sqrt{x}$$

007: sqrt_group

$$\sqrt{a+b}$$

008: sqrt_index_paren

$$\sqrt[(3)]{x}$$

009: sqrt_index_brace

$$\sqrt[3]{x}$$

010: root_ident

$$\sqrt{x}$$

011: root_group

$$\sqrt{a+b}$$

012: root_index

$$\sqrt[(3)]{x}$$

013: matrix_2x2

$$a\ b$$
$$c\ d$$

014: matrix_col

$$1$$
$$2$$
$$3$$

015: matrix_row

$$a\ b\ c$$

016: matrix_no_brace

017: pmatrix_2x2

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

018: pmatrix_1x1

$$\begin{pmatrix} x \end{pmatrix}$$

019: pmatrix_frac

$$\begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{pmatrix}$$

020: bmatrix_2x2

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

021: bmatrix_identity

$$\begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}$$

022: bmatrix_1x2

$$\begin{bmatrix}x & y\end{bmatrix}$$

023: dmatrix_2x2

$$\begin{vmatrix}a & b\\c & d\end{vmatrix}$$

024: dmatrix_col

$$\begin{vmatrix}x\\y\end{vmatrix}$$

025: dmatrix_1x1

$$\begin{vmatrix} 1 \end{vmatrix}$$

026: eqalign_two

$$\begin{aligned} x &= 1 \\ y &= 2 \end{aligned}$$

027: eqalign_one

$$a + b = c$$

028: eqalign_no_amp

$$\begin{aligned} x &= 1 \\ y &= 2 \end{aligned}$$

029: pile_center

a
b
c

030: lpile_left

a
b

031: rpile_right

a
b

032: over_bare

033: over_empty_numer

034: over_empty_denom

$$\frac{1}{}$$

035: over_lower

$$\frac{1}{2}$$

036: over_mixed_case

$$\frac{1}{2}$$

037: over_chain_three

$$\frac{\frac{1}{2}}{3}$$

038: over_chain_four

$$\frac{\frac{\frac{1}{2}}{3}}{4}$$

039: over_ws_pad

$$\frac{1}{2}$$

040: over_idents

$$\frac{a}{b}$$

041: over_greek

$$\frac{A}{B}$$

042: over_func

$$\sin\frac{x}{\cos}x$$

043: over_sqrt

$$\frac{\sqrt{2}}{\sqrt{3}}$$

044: over_empty_groups

$$\frac{}{}$$

045: over_glued_both

$$\frac{1}{2}$$

046: over_neg

$$\frac{-1}{-2}$$

047: over_decimal

$$\frac{1.5}{2.5}$$

048: over_multidigit

$$\frac{123}{456}$$

049: over_sup

$$\frac{x^2}{y^2}$$

050: over_sub

$$\frac{x_1}{y_1}$$

051: over_subsup

$$\frac{x_i^2}{y_j^3}$$

052: over_triple_nest

$$\cfrac{\cfrac{1}{2}}{\cfrac{3}{4}}$$

053: over_in_left_brace

$$\left\{\frac{a}{b}\right\}$$

054: over_sum

$$\sum_{i}\frac{a_i}{n}$$

055: over_matrix_numer

$$\frac{\begin{matrix}1 & 2\\ 3 & 4\end{matrix}}{2}$$

056: over_plus_ungrouped

$$\frac{a+b}{c+d}$$

057: over_paren_no_left

$$\frac{(x}{y)}$$

058: over_times

$$a \times \frac{b}{c} \times d$$

059: over_pi

$$\frac{\pi}{2}$$

060: over_infty

$$\frac{1}{\infty}$$

061: atop_simple

$$\begin{matrix}1\\2\end{matrix}$$

062: atop_grouped

$$\begin{matrix}a+b\\c-d\end{matrix}$$

063: atop_chain

$$\begin{matrix}1\\2\\3\end{matrix}$$

064: frac_latex

$FRACC12$

065: dfrac_latex

$$DFRAC12$$

066: tfrac_latex

$$TFRACab$$

067: frac_missing_denom

$$FRAC1$$

068: over_root

$$\frac{\sqrt{8}}{\sqrt{2}}$$

069: sqrt_bare

$$\sqrt{}$$

070: root_bare

$$\sqrt{}$$

071: sqrt_lower

$$\sqrt{x}$$

072: root_lower

$$\sqrt{x}$$

073: sqrt_latex_index

$$\sqrt[3]{x}$$

074: sqrt_latex_empty_idx

$$\sqrt[]{x}$$

075: sqrt_paren_no_of

$$\sqrt[3]{x}$$

076: sqrt_empty_paren

$$\sqrt[()]{x}$$

077: root_brace_of

$$\sqrt[3]{x}$$

078: root_latex_index

$$\sqrt{[4]}\,y$$

079: sqrt_nested

$$\sqrt{\sqrt{x}}$$

080: sqrt_of_frac

$$\sqrt{\frac{1}{2}}$$

081: root_of_frac

$$\sqrt{\frac{a}{b}}$$

082: sqrt_of_matrix

$$\sqrt{\begin{matrix}1 & 0\\ 0 & 1\end{matrix}}$$

083: sqrt_sup

$$\sqrt{x^2}$$

084: sqrt_empty_body

$$\sqrt{}$$

085: root_empty_body

$\sqrt{}$

086: sqrt_long

$\sqrt{a+b+c+d+e+f}$

087: sqrt_mixed_case

$SqRtx$

088: root_mixed_case

$RoOta+b$

089: sqrt_of_pile

$$\sqrt{\frac{a}{b}}$$

090: root_of_paren_idx

$$\sqrt[(n)]{x+y}$$

091: sqrt_index_zero

$$\sqrt[(0)]{1}$$

092: sqrt_index_frac

$$\sqrt[\frac{1}{2}]{x}$$

093: root_index_frac

$$\sqrt[\frac{1}{2}]{x}$$

094: sqrt_of_over_chain

$$\sqrt{\frac{1}{\frac{2}{3}}}$$

095: root_nested

$$\sqrt{\sqrt{x}}$$

096: sqrt_int

$$\sqrt{\int_{0}^{1} x dx}$$

097: sqrt_of_pmatrix

$$\sqrt{\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}}$$

098: matrix_empty_brace

099: matrix_1x1

$$z$$

100: matrix_lower

$$\begin{matrix} a & b \\ c & d \end{matrix}$$

101: matrix_2x3

$$\begin{matrix} a & b & c \\ d & e & f \end{matrix}$$

102: matrix_3x2

$$\begin{matrix} a & b \\ c & d \\ e & f \end{matrix}$$

103: matrix_3x3

$$\begin{matrix} a & b & c \\ d & e & f \\ g & h & i \end{matrix}$$

104: matrix_trailing_hash

$$\begin{matrix} a & b \end{matrix}$$

105: matrix_trailing_amp

$$a \; b$$

106: matrix_jagged

$$\begin{matrix} a \; b \; c \\ d \; e \; f \end{matrix}$$

107: matrix_spacey

$$\begin{matrix} a \; b \\ c \; d \end{matrix}$$

108: pmatrix_empty_brace

$$\begin{pmatrix} \end{pmatrix}$$

109: pmatrix_no_brace

$$\begin{pmatrix} \end{pmatrix}$$

110: pmatrix_lower

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

111: pmatrix_2x3

$$\begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix}$$

112: pmatrix_3x2

$$\begin{pmatrix} a & b \\ c & d \\ e & f \end{pmatrix}$$

113: pmatrix_3x3

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$$

114: pmatrix_trailing_hash

$$\begin{pmatrix} a & b \\ \end{pmatrix}$$

115: pmatrix_trailing_amp

$$\begin{pmatrix} a & b & \end{pmatrix}$$

116: pmatrix_jagged

$$\begin{pmatrix} a & b & c \\ d & e & f \\ \end{pmatrix}$$

117: pmatrix_spacey

$$\begin{pmatrix} a\ b \\ c\ d \end{pmatrix}$$

118: bmatrix_empty_brace

$$\begin{bmatrix} \end{bmatrix}$$

119: bmatrix_no_brace

$$\begin{bmatrix} \end{bmatrix}$$

120: bmatrix_1x1

$$\begin{bmatrix} z \end{bmatrix}$$

121: bmatrix_lower

$$\begin{bmatrix} a\ b \\ c\ d \end{bmatrix}$$

122: bmatrix_2x3

$$\begin{bmatrix} a\ b\ c \\ d\ e\ f \end{bmatrix}$$

123: bmatrix_3x2

$$\begin{bmatrix} a\ b \\ c\ d \\ e\ f \end{bmatrix}$$

124: bmatrix_3x3

$$\begin{bmatrix} a\ b\ c \\ d\ e\ f \\ g\ h\ i \end{bmatrix}$$

125: bmatrix_trailing_hash

$$\begin{bmatrix} a & b \\ & \end{bmatrix}$$

126: bmatrix_trailing_amp

$$\begin{bmatrix} a & b & \end{bmatrix}$$

127: bmatrix_jagged

$$\begin{bmatrix} a & b & c \\ d & e & f \\ & & \end{bmatrix}$$

128: bmatrix_spacey

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

129: dmatrix_empty_brace

$$\begin{vmatrix} \end{vmatrix}$$

130: dmatrix_no_brace

$$\begin{vmatrix} \end{vmatrix}$$

131: dmatrix_lower

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

132: dmatrix_2x3

$$\begin{vmatrix} a & b & c \\ d & e & f \end{vmatrix}$$

133: dmatrix_3x2

$$\begin{vmatrix} a & b \\ c & d \\ e & f \end{vmatrix}$$

134: dmatrix_3x3

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}$$

135: dmatrix_trailing_hash

$$\begin{vmatrix} a & b \\ & \end{vmatrix}$$

136: dmatrix_trailing_amp

$$\begin{vmatrix} a & b \end{vmatrix}$$

137: dmatrix_jagged

$$\left|\begin{matrix} a & b & c \\ d & e & f \end{matrix}\right|$$

138: dmatrix_spacey

$$\left|\begin{matrix} a & b \\ c & d \end{matrix}\right|$$

139: matrix_empty_row

$$\begin{matrix} a & b \\ & c \\ d \end{matrix}$$

140: matrix_amp_only

141: matrix_hash_only

142: matrix_sqrt_cells

$$\begin{matrix}\sqrt{1} & \sqrt{2} \\ \sqrt{3} & \sqrt{4}\end{matrix}$$

143: pmatrix_over_all

$$\begin{pmatrix}\frac{1}{2} & \frac{3}{4} \\ \frac{5}{6} & \frac{7}{8}\end{pmatrix}$$

144: bmatrix_func

$$\begin{bmatrix}\sin x & \cos x \\ -\sin x & \cos x\end{bmatrix}$$

145: dmatrix_nested_over

$$\begin{vmatrix} \frac{1}{2} & 0 \\ \frac{3}{0} & 1 \end{vmatrix}$$

146: vmatrix_2x2

$$VMATRIXab$$

147: smallmatrix_2x2

$$SMALLMATRIXab$$

148: vmatrix_no_brace

$$VMATRIX$$

149: smallmatrix_no_brace

$$\mathit{SMALLMATRIX}$$

150: ladder_2x2

$$a \left\lfloor \underline{b} \right. \\ \quad c$$

151: sladder_2x2

$$a \left\lfloor \underline{b} \right. \\ \quad c$$

152: ladder_no_brace

153: sladder_no_brace

154: matrix_over_3x3

$$\begin{matrix} \frac{1}{2} & \frac{3}{4} & \frac{5}{6} \\ \frac{7}{8} & \frac{9}{10} & \frac{11}{12} \\ \frac{13}{14} & \frac{15}{16} & \frac{17}{18} \end{matrix}$$

155: matrix_5x5_over

$$\begin{matrix} \frac{1}{2} & \frac{3}{4} & \frac{5}{6} & \frac{7}{8} & \frac{9}{10} \\ \frac{11}{12} & \frac{13}{14} & \frac{15}{16} & \frac{17}{18} & \frac{19}{20} \\ \frac{21}{22} & \frac{23}{24} & \frac{25}{26} & \frac{27}{28} & \frac{29}{30} \\ \frac{31}{32} & \frac{33}{34} & \frac{35}{36} & \frac{37}{38} & \frac{39}{40} \\ \frac{41}{42} & \frac{43}{44} & \frac{45}{46} & \frac{47}{48} & \frac{49}{50} \end{matrix}$$

156: matrix_5x5_sqrt

$$\begin{matrix} \sqrt[2]{5} & \sqrt[3]{6} & \sqrt[4]{7} & \sqrt[5]{8} & \sqrt[6]{9} \\ \sqrt[7]{10} & \sqrt[8]{11} & \sqrt[9]{12} & \sqrt[10]{13} & \sqrt[11]{14} \\ \sqrt[12]{15} & \sqrt[13]{16} & \sqrt[14]{17} & \sqrt[15]{18} & \sqrt[16]{19} \\ \sqrt[17]{20} & \sqrt[18]{21} & \sqrt[19]{22} & \sqrt[20]{23} & \sqrt[21]{24} \\ \sqrt[22]{25} & \sqrt[23]{26} & \sqrt[24]{27} & \sqrt[25]{28} & \sqrt[26]{29} \end{matrix}$$

157: matrix_4x4_mixed

$$\begin{matrix} 1 & \frac{1}{3} & \sqrt[(2)]{7} & d \\ \frac{2}{2} & \sqrt[(3)]{6} & d & 8 \\ \sqrt[(4)]{5} & d & 11 & \frac{3}{5} \\ d & 14 & \frac{4}{4} & \sqrt[(5)]{8} \end{matrix}$$

158: matrix_4x4_numbers

$$\begin{matrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{matrix}$$

159: matrix_3x3_over

$$\begin{matrix} \frac{21}{22} & \frac{23}{24} & \frac{25}{26} \\ \frac{27}{28} & \frac{29}{30} & \frac{31}{32} \\ \frac{33}{34} & \frac{35}{36} & \frac{37}{38} \end{matrix}$$

160: pmatrix_5x5_over

$$\begin{pmatrix} \frac{1}{2} & \frac{3}{4} & \frac{5}{6} & \frac{7}{8} & \frac{9}{10} \\ \frac{11}{12} & \frac{13}{14} & \frac{15}{16} & \frac{17}{18} & \frac{19}{20} \\ \frac{21}{22} & \frac{23}{24} & \frac{25}{26} & \frac{27}{28} & \frac{29}{30} \\ \frac{31}{32} & \frac{33}{34} & \frac{35}{36} & \frac{37}{38} & \frac{39}{40} \\ \frac{41}{42} & \frac{43}{44} & \frac{45}{46} & \frac{47}{48} & \frac{49}{50} \end{pmatrix}$$

161: pmatrix_5x5_sqrt

$$\begin{pmatrix} \sqrt[2]{5} & \sqrt[3]{6} & \sqrt[4]{7} & \sqrt[5]{8} & \sqrt[6]{9} \\ \sqrt[7]{10} & \sqrt[8]{11} & \sqrt[9]{12} & \sqrt[10]{13} & \sqrt[11]{14} \\ \sqrt[12]{15} & \sqrt[13]{16} & \sqrt[14]{17} & \sqrt[15]{18} & \sqrt[16]{19} \\ \sqrt[17]{20} & \sqrt[18]{21} & \sqrt[19]{22} & \sqrt[20]{23} & \sqrt[21]{24} \\ \sqrt[22]{25} & \sqrt[23]{26} & \sqrt[24]{27} & \sqrt[25]{28} & \sqrt[26]{29} \end{pmatrix}$$

162: pmatrix_4x4_mixed

$$\begin{pmatrix} 1 & \frac{1}{3} & \sqrt[2]{7} & d \\ \frac{2}{2} & \sqrt[3]{6} & d & 8 \\ \sqrt[4]{5} & d & 11 & \frac{3}{5} \\ d & 14 & \frac{4}{4} & \sqrt[5]{8} \end{pmatrix}$$

163: pmatrix_4x4_numbers

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{pmatrix}$$

164: pmatrix_3x3_over

$$\begin{pmatrix} \frac{21}{22} & \frac{23}{24} & \frac{25}{26} \\ \frac{27}{28} & \frac{29}{30} & \frac{31}{32} \\ \frac{33}{34} & \frac{35}{36} & \frac{37}{38} \end{pmatrix}$$

165: bmatrix_5x5_over

$$\begin{bmatrix} \frac{1}{2} & \frac{3}{4} & \frac{5}{6} & \frac{7}{8} & \frac{9}{10} \\ \frac{11}{12} & \frac{13}{14} & \frac{15}{16} & \frac{17}{18} & \frac{19}{20} \\ \frac{21}{22} & \frac{23}{24} & \frac{25}{26} & \frac{27}{28} & \frac{29}{30} \\ \frac{31}{32} & \frac{33}{34} & \frac{35}{36} & \frac{37}{38} & \frac{39}{40} \\ \frac{41}{42} & \frac{43}{44} & \frac{45}{46} & \frac{47}{48} & \frac{49}{50} \end{bmatrix}$$

166: bmatrix_5x5_sqrt

$$\begin{bmatrix} \sqrt[2]{5} & \sqrt[3]{6} & \sqrt[4]{7} & \sqrt[5]{8} & \sqrt[6]{9} \\ \sqrt[7]{10} & \sqrt[8]{11} & \sqrt[9]{12} & \sqrt[10]{13} & \sqrt[11]{14} \\ \sqrt[12]{15} & \sqrt[13]{16} & \sqrt[14]{17} & \sqrt[15]{18} & \sqrt[16]{19} \\ \sqrt[17]{20} & \sqrt[18]{21} & \sqrt[19]{22} & \sqrt[20]{23} & \sqrt[21]{24} \\ \sqrt[22]{25} & \sqrt[23]{26} & \sqrt[24]{27} & \sqrt[25]{28} & \sqrt[26]{29} \end{bmatrix}$$

167: bmatrix_4x4_mixed

$$\begin{bmatrix} 1 & \frac{1}{3} & \sqrt[2]{7} & d \\ \frac{2}{2} & \sqrt[3]{6} & d & 8 \\ \sqrt[4]{5} & d & 11 & \frac{3}{5} \\ d & 14 & \frac{4}{4} & \sqrt[5]{8} \end{bmatrix}$$

168: bmatrix_4x4_numbers

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{bmatrix}$$

169: bmatrix_3x3_over

$$\begin{bmatrix} \frac{21}{22} & \frac{23}{24} & \frac{25}{26} \\ \frac{27}{28} & \frac{29}{30} & \frac{31}{32} \\ \frac{33}{34} & \frac{35}{36} & \frac{37}{38} \end{bmatrix}$$

170: dmatrix_5x5_over

$$\begin{vmatrix} \frac{1}{2} & \frac{3}{4} & \frac{5}{6} & \frac{7}{8} & \frac{9}{10} \\ \frac{11}{12} & \frac{13}{14} & \frac{15}{16} & \frac{17}{18} & \frac{19}{20} \\ \frac{21}{22} & \frac{23}{24} & \frac{25}{26} & \frac{27}{28} & \frac{29}{30} \\ \frac{31}{32} & \frac{33}{34} & \frac{35}{36} & \frac{37}{38} & \frac{39}{40} \\ \frac{41}{42} & \frac{43}{44} & \frac{45}{46} & \frac{47}{48} & \frac{49}{50} \end{vmatrix}$$

171: dmatrix_5x5_sqrt

$$\begin{vmatrix} \sqrt[2]{5} & \sqrt[3]{6} & \sqrt[4]{7} & \sqrt[5]{8} & \sqrt[6]{9} \\ \sqrt[7]{10} & \sqrt[8]{11} & \sqrt[9]{12} & \sqrt[10]{13} & \sqrt[11]{14} \\ \sqrt[12]{15} & \sqrt[13]{16} & \sqrt[14]{17} & \sqrt[15]{18} & \sqrt[16]{19} \\ \sqrt[17]{20} & \sqrt[18]{21} & \sqrt[19]{22} & \sqrt[20]{23} & \sqrt[21]{24} \\ \sqrt[22]{25} & \sqrt[23]{26} & \sqrt[24]{27} & \sqrt[25]{28} & \sqrt[26]{29} \end{vmatrix}$$

172: dmatrix_4x4_mixed

$$\begin{vmatrix} 1 & \frac{1}{3} & \sqrt[2]{7} & d \\ \frac{2}{2} & \sqrt[3]{6} & d & 8 \\ \sqrt[4]{5} & d & 11 & \frac{3}{5} \\ d & 14 & \frac{4}{4} & \sqrt[5]{8} \end{vmatrix}$$

173: dmatrix_4x4_numbers

$$\left|\begin{matrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{matrix}\right|$$

174: dmatrix_3x3_over

$$\left|\begin{matrix} \frac{21}{22} & \frac{23}{24} & \frac{25}{26} \\ \frac{27}{28} & \frac{29}{30} & \frac{31}{32} \\ \frac{33}{34} & \frac{35}{36} & \frac{37}{38} \end{matrix}\right|$$

175: matrix_7x3_over

$$\begin{matrix} \frac{41}{42} & \frac{43}{44} & \frac{45}{46} \\ \frac{47}{48} & \frac{49}{50} & \frac{51}{52} \\ \frac{53}{54} & \frac{55}{56} & \frac{57}{58} \\ \frac{59}{60} & \frac{61}{62} & \frac{63}{64} \\ \frac{65}{66} & \frac{67}{68} & \frac{69}{70} \\ \frac{71}{72} & \frac{73}{74} & \frac{75}{76} \\ \frac{77}{78} & \frac{79}{80} & \frac{81}{82} \end{matrix}$$

176: matrix_3x7_over

$$\begin{matrix} \frac{81}{82} & \frac{83}{84} & \frac{85}{86} & \frac{87}{88} & \frac{89}{90} & \frac{91}{92} & \frac{93}{94} \\ \frac{95}{96} & \frac{97}{98} & \frac{99}{100} & \frac{101}{102} & \frac{103}{104} & \frac{105}{106} & \frac{107}{108} \\ \frac{109}{110} & \frac{111}{112} & \frac{113}{114} & \frac{115}{116} & \frac{117}{118} & \frac{119}{120} & \frac{121}{122} \end{matrix}$$

177: pmatrix_7x3_over

$$\left(\begin{array}{ccc} 41 & 43 & 45 \\ \hline 42 & 44 & 46 \\ \hline 47 & 49 & 51 \\ \hline 48 & 50 & 52 \\ \hline 53 & 55 & 57 \\ \hline 54 & 56 & 58 \\ \hline 59 & 61 & 63 \\ \hline 60 & 62 & 64 \\ \hline 65 & 67 & 69 \\ \hline 66 & 68 & 70 \\ \hline 71 & 73 & 75 \\ \hline 72 & 74 & 76 \\ \hline 77 & 79 & 81 \\ \hline 78 & 80 & 82 \end{array}\right)$$

178: pmatrix_3x7_over

$$\left(\begin{array}{ccccccc} 81 & 83 & 85 & 87 & 89 & 91 & 93 \\ \hline 82 & 84 & 86 & 88 & 90 & 92 & 94 \\ \hline 95 & 97 & 99 & 101 & 103 & 105 & 107 \\ \hline 96 & 98 & 100 & 102 & 104 & 106 & 108 \\ \hline 109 & 111 & 113 & 115 & 117 & 119 & 121 \\ \hline 110 & 112 & 114 & 116 & 118 & 120 & 122 \end{array}\right)$$

179: bmatrix_7x3_over

$$\left[\begin{array}{ccc} 41 & 43 & 45 \\ \hline 42 & 44 & 46 \\ \hline 47 & 49 & 51 \\ \hline 48 & 50 & 52 \\ \hline 53 & 55 & 57 \\ \hline 54 & 56 & 58 \\ \hline 59 & 61 & 63 \\ \hline 60 & 62 & 64 \\ \hline 65 & 67 & 69 \\ \hline 66 & 68 & 70 \\ \hline 71 & 73 & 75 \\ \hline 72 & 74 & 76 \\ \hline 77 & 79 & 81 \\ \hline 78 & 80 & 82 \end{array}\right]$$

180: bmatrix_3x7_over

$$\left[\begin{array}{ccccccc} 81 & 83 & 85 & 87 & 89 & 91 & 93 \\ \hline 82 & 84 & 86 & 88 & 90 & 92 & 94 \\ \hline 95 & 97 & 99 & 101 & 103 & 105 & 107 \\ \hline 96 & 98 & 100 & 102 & 104 & 106 & 108 \\ \hline 109 & 111 & 113 & 115 & 117 & 119 & 121 \\ \hline 110 & 112 & 114 & 116 & 118 & 120 & 122 \end{array}\right]$$

181: dmatrix_7x3_over

$$\begin{array}{ccc} 41 & 43 & 45 \\ \hline 42 & 44 & 46 \\ \hline 47 & 49 & 51 \\ \hline 48 & 50 & 52 \\ \hline 53 & 55 & 57 \\ \hline 54 & 56 & 58 \\ \hline 59 & 61 & 63 \\ \hline 60 & 62 & 64 \\ \hline 65 & 67 & 69 \\ \hline 66 & 68 & 70 \\ \hline 71 & 73 & 75 \\ \hline 72 & 74 & 76 \\ \hline 77 & 79 & 81 \\ \hline 78 & 80 & 82 \end{array}$$

182: dmatrix_3x7_over

$$\left|\begin{array}{ccccccc} 81 & 83 & 85 & 87 & 89 & 91 & 93 \\ \hline 82 & 84 & 86 & 88 & 90 & 92 & 94 \\ \hline 95 & 97 & 99 & 101 & 103 & 105 & 107 \\ \hline 96 & 98 & 100 & 102 & 104 & 106 & 108 \\ \hline 109 & 111 & 113 & 115 & 117 & 119 & 121 \\ \hline 110 & 112 & 114 & 116 & 118 & 120 & 122 \end{array}\right|$$

183: eqalign_no_brace

184: eqalign_empty

185: eqalign_three

$$x = 1$$
$$y = 2$$
$$z = 3$$

186: eqalign_four

$$a = 1$$
$$b = 2$$
$$c = 3$$
$$d = 4$$

187: eqalign_left_only

$$x$$
$$y$$

188: eqalign_andand

$$x1$$

189: eqalign_over

$$\frac{a}{b} = \frac{c}{d}$$

190: eqalign_sqrt

$$\sqrt{x} = 2$$
$$\sqrt{y} = 3$$

191: eqalign_trailing_hash

$$x = 1$$

192: eqalign_lower

$$x = 1$$
$$y = 2$$

193: eqalign_mixed_amp

$$x = 1$$
$$y = 2$$

194: eqalign_six_over

$$\frac{1}{2} = \frac{3}{4}$$
$$\frac{2}{3} = \frac{4}{5}$$
$$\frac{3}{4} = \frac{5}{6}$$
$$\frac{4}{5} = \frac{6}{7}$$
$$\frac{5}{6} = \frac{7}{8}$$
$$\frac{6}{7} = \frac{8}{9}$$

195: eqalign_ten_over

$$\frac{1}{2} = \frac{3}{4}$$
$$\frac{2}{3} = \frac{4}{5}$$
$$\frac{3}{4} = \frac{5}{6}$$
$$\frac{4}{5} = \frac{6}{7}$$
$$\frac{5}{6} = \frac{7}{8}$$
$$\frac{6}{7} = \frac{8}{9}$$
$$\frac{7}{8} = \frac{9}{10}$$
$$\frac{8}{9} = \frac{10}{11}$$
$$\frac{9}{10} = \frac{11}{12}$$
$$\frac{10}{11} = \frac{12}{13}$$

196: eqalign_matrix

$$A = \begin{matrix} 1 & 0 \\ 0 & 1 \end{matrix}$$
$$B = \begin{matrix} 0 & 1 \\ 1 & 0 \end{matrix}$$

197: eqalign_root

$$\sqrt{8} = 2$$
$$\sqrt{27} = 3$$

198: pile_no_brace

199: lpile_no_brace

200: rpile_no_brace

201: pile_empty

202: pile_single

a

203: pile_over

$\frac{1}{2}$

$\frac{3}{4}$

204: pile_sqrt

$\sqrt{a}$

$\sqrt{b}$

$\sqrt{c}$

205: pile_five

$$a$$
$$b$$
$$c$$
$$d$$
$$e$$

206: pile_lower

$$a$$
$$b$$

207: lpile_over

$$\frac{1}{2}$$
$$x$$

208: rpile_over

$$\frac{1}{2}$$
$$x$$

209: pile_eight_frac

$$\frac{1}{2}$$
$$\frac{2}{3}$$
$$\frac{3}{4}$$
$$\frac{4}{5}$$
$$\frac{5}{6}$$
$$\frac{6}{7}$$
$$\frac{7}{8}$$
$$\frac{8}{9}$$

210: lpile_eight_frac

$$\frac{1}{2}$$
$$\frac{2}{3}$$
$$\frac{3}{4}$$
$$\frac{4}{5}$$
$$\frac{5}{6}$$
$$\frac{6}{7}$$
$$\frac{7}{8}$$
$$\frac{8}{9}$$

211: rpile_eight_frac

$$\frac{1}{2}$$
$$\frac{2}{3}$$
$$\frac{3}{4}$$
$$\frac{4}{5}$$
$$\frac{5}{6}$$
$$\frac{6}{7}$$
$$\frac{7}{8}$$
$$\frac{8}{9}$$

212: lpile_three

a
b
c

213: rpile_three

$$
\begin{array}{c}
a \\
b \\
c
\end{array}
$$

214: pile_matrix

$$
\begin{array}{cc}
1 & 0 \\
0 & 1 \\
0 & 1 \\
1 & 0
\end{array}
$$

215: lpile_lower

$$
\begin{array}{l}
a \\
b
\end{array}
$$

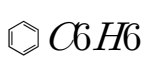
216: rpile_lower

$$
\begin{array}{r}
a \\
b
\end{array}
$$

217: benzene



218: benzene_group



219: bigg_paren

220: bigg_ident



221: choose_simple

$$\binom{n}{k}$$

222: binom_simple

$$\binom{n}{k}$$

223: longdiv_simple

$$3\overline{)6}^{\,2}$$

224: color_red

$$x$$

225: unknown_foobar

$$FOOBAR$$

226: unknown_xyz

$$XYZ\backslash 1$$

227: cases_2

$$\begin{cases} x & \text{if} x > 0 \\ -x \text{if} x < 0 \end{cases}$$

228: cases_no_brace

$$\{$$

229: rel_arrow

$$\xrightarrow[g]{f}$$

230: buildrel_arrow

$$\xrightarrow{\text{def}}$$

231: phantom_x

232: begin_matrix

$$\mathit{BEGIN}$$

233: end_matrix_alone

$$END$$

234: begin_pmatrix

$$BEGIN(\ )$$

235: quad_space

$$aQUADb$$

236: operatorname_sin

$$OPERATORNAME\sin$$

237: text_cmd

$$TEXTif$$

238: overset_hat

239: underset_bar

$$\underline{SET}$$

240: stackrel_star

$$STACKREL{*}{=}$$

241: lim_cmd

$$\lim_{x \to 0} f(x)$$

242: lim_upper_cmd

$$\operatorname{Lim}_{n \to \infty} a_n$$

243: lsub_cmd

244: lsup_cmd

245: sub_cmd

246: sup_cmd

247: over_eqalign

$$\frac{1}{2} = \frac{3}{6}$$

248: over_pile

$$\frac{\frac{1}{2}}{\frac{\frac{3}{4}}{5}}$$

249: sqrt_eqalign

$$\sqrt{x} = y$$

250: root_matrix

$$\sqrt{\begin{matrix} 1 & 0 \\ 0 & 1 \end{matrix}}$$

251: over_choose

$$\frac{\binom{n}{k}}{2}$$

252: matrix_cases_cell

$$\begin{cases} 1 \\ 0 & 2 \end{cases}$$

253: pmatrix_pile_cell

$$\binom{a}{b}$$

254: bmatrix_eqalign_cell

$$\begin{bmatrix} x = 1 \\ 0 \quad 0 \end{bmatrix} 1$$

255: dmatrix_root_cell

$$\begin{vmatrix} \sqrt{2} & 0 \\ 0 & \sqrt{3} \end{vmatrix}$$

256: over_nested_matrix_3

$$\frac{\begin{matrix} \frac{1}{2} & \frac{3}{4} \\ \frac{5}{6} & \frac{7}{8} \end{matrix}}{\begin{matrix} \frac{9}{10} & \frac{11}{12} \\ \frac{13}{14} & \frac{15}{16} \end{matrix}}$$

257: sqrt_nested_pmatrix_3

$$\sqrt[3]{\begin{pmatrix} \frac{1}{2} & \frac{3}{4} \\ \frac{5}{6} & \frac{7}{8} \end{pmatrix}}$$

258: eqalign_matrix_fracs

$$\begin{matrix} \frac{1}{2} & 0 \\ 0 & \frac{3}{4} \end{matrix} = I$$

$$\left.\begin{pmatrix} \frac{5}{6} & 1 \\ 1 & \frac{7}{8} \end{pmatrix}\right. = A$$

259: pile_four_matrices

$$\begin{matrix} 1\,0 \\ 0\,1 \end{matrix}$$

$$\begin{pmatrix} 1\,0 \\ 0\,1 \end{pmatrix}$$

$$\begin{bmatrix} 1\,0 \\ 0\,1 \end{bmatrix}$$

$$\begin{vmatrix} 1\,0 \\ 0\,1 \end{vmatrix}$$